

DSONet: A Lightweight Framework for the Detection of Multiscale Dense Ship Occlusion

Chang Qu^{1b}, Yuhu Shi^{1b}, Hongyu Chen^{1b}, Qianqian Ye^{1b}, Yang Cai^{1b}, Yugang Chang^{1b},
Chengcheng Chen^{1b}, *Graduate Student Member, IEEE*, Fei Wang^{1b}, and Weiming Zeng^{1b}, *Senior Member, IEEE*

Abstract—With the rise of autonomous shipping and intelligent maritime surveillance, accurately detecting ships under multiscale dense occlusions has become a critical challenge. Traditional detectors struggle with feature degradation caused by overlapping targets and scale variation. To address this, we propose DSONet, a lightweight framework tailored for robust ship detection in complex maritime environments. Specifically, we design a task-specific feature extraction structure, DualFeatureDetection, which enhances spatial detail preservation while reducing redundant computation. Additionally, we introduce the AdaptiveUpsample (AU) module to improve multiscale feature fusion and spatial reconstruction, especially under occlusion. Integrated with a four-branch oriented bounding box (OBB) detection head, DSONet achieves precise localization of elongated and overlapping ships. Extensive experiments on the MID, SeaShip and SSDD datasets demonstrate that DSONet outperforms existing detectors in both accuracy and efficiency, offering a practical solution for maritime occlusion detection.

Index Terms—Lightweight model, multiscale detection, ship occlusion detection, target detection,

I. INTRODUCTION

WITH the rapid growth of global maritime transport, the density of ships in ports and waterways has steadily increased, making ship occlusion detection a crucial technology in intelligent maritime monitoring [1] and autonomous shipping systems [2]. Accurate detection and tracking of ships is essential for ensuring navigation safety, optimizing port logistics, and enhancing automation in maritime traffic management [3], emergency rescue operations [4], and intelligent shipping systems [5]. However, two major challenges persist: multiscale object variation [6] and dense target occlusion.

In practical maritime scenes, ships frequently exhibit behaviors, such as convoy sailing, maneuvering, and close docking, which lead to frequent overlaps and severe occlusions [7]. Moreover, ship sizes vary dramatically—from small nearshore fishing boats to large offshore cargo vessels—posing difficulties for traditional single-scale detectors. Compounding

this, complex oceanic conditions with varying wave patterns, lighting, and weather further degrade detection performance, requiring algorithms with strong spatial awareness and adaptability.

Deep learning-based detectors have greatly advanced object detection. Two-stage methods, such as Faster R-CNN [8], utilize a region proposal network for high-accuracy localization, but are computationally intensive and ill-suited for real-time applications. Single-stage detectors like YOLO [9] and SSD [10] enable faster inference through end-to-end regression, but struggle with small object detection and densely crowded scenes. YOLOv8, YOLOv9 [11], and YOLOv10 [12] have improved in accuracy and scale sensitivity through backbone refinement and loss optimization. EfficientDet [13] proposes BiFPN for better multiscale fusion while reducing computational cost, facilitating lightweight deployment in constrained environments. Transformer-based architectures [14] have recently gained popularity in object detection by modeling long-range dependencies and global context. For instance, DETR [15] captures global relationships through self-attention mechanisms, eliminating reliance on local convolutional features. However, DETR suffers from slow convergence. Extensions, such as Deformable DETR [16] and RT-DETR [17] accelerate training and inference by combining deformable attention with CNNs. These models show improved flexibility but often remain resource-heavy for edge applications.

Furthermore, some studies have proposed novel methods to tackle dense occlusion detection. Liang et al. introduced CBNet [18], which uses a dual-backbone network structure to strengthen feature extraction and improve detection performance in occluded scenarios. Cao et al. [19] developed Pisa, optimizing positive and negative sample allocation strategies to enhance the model's discrimination ability in occlusion situations. Additionally, Law and Deng's CornerNet [20] locates object corners through corner point detection, effectively handling object overlap and occlusion, thus improving detection accuracy in complex scenes.

While recent methods have improved detection accuracy, they often entail high computational costs, limiting their applicability in real-time or resource-constrained environments. Notably, there is a growing demand for deploying ship detection systems on edge platforms—such as autonomous aerial vehicleless (AAVs), autonomous surface vessels (ASVs), and shipboard terminals—where power efficiency, inference latency, and model size become critical constraints. Contrary

Received 5 May 2025; revised 12 July 2025; accepted 26 July 2025. Date of publication 30 July 2025; date of current version 9 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61906117, and in part by the 2023 Graduate Top Innovative Talents Training Program at Shanghai Maritime University under Grant 2023YBR013. (Corresponding author: Yuhu Shi.)

The authors are with the Information Engineering College, Shanghai Maritime University, Shanghai 201306, China (e-mail: 13347039265@163.com; syhustb2011@163.com; hongychen676@gmail.com; 1759039741@qq.com; icyasd333@gmail.com; cygang_post@163.com; shmtu_ccc@163.com; shine_wxf@163.com; zengwm86@163.com).

Digital Object Identifier 10.1109/IIOT.2025.3593829

to the assumption that maritime detection always relies on high-performance workstations, many practical applications, including mobile surveillance and onboard autonomy, require lightweight yet robust solutions. Addressing this gap, we propose DSONet, a compact and efficient framework tailored for multiscale dense ship occlusion scenarios. By balancing detection accuracy with deployment feasibility, DSONet enables reliable maritime perception under real-world operational constraints.

The main contributions of this study are summarized as follows.

- 1) A feature extraction network, the DualFeature-Detection (DF-Det) structure, employs the designed DualFeatureConv (DFC) module. This module enhances multiscale feature representation through dual channel feature extraction, improving detection accuracy in complex maritime environments. By incorporating the DualFusion (DF) structure, the overall structure optimizes feature learning for dense targets while maintaining efficient computation, effectively reducing detection errors caused by occlusion and scale variation.
- 2) A content-aware adaptive upsampling module AdaptiveUpsample (AU) is designed. This module employs a dynamic sampling weight mechanism to adaptively adjust the feature reorganization process, optimize multiscale feature fusion, and enhance the detection capability for densely occluded targets.
- 3) We propose DSONet, a lightweight framework specifically designed for multiscale dense ship occlusion detection. By integrating the AU module into the DF-Det framework, DSONet improves both detection accuracy and computational efficiency.

The remainder of this article is organized as follows. Section II reviews related work on occlusion detection. The proposed DSONet framework is presented in Section III. Experimental results and comparisons are discussed in Section IV. Finally, discussions and conclusions are drawn in Sections V and VI.

II. RELATED WORKS

A. Occlusion Detection Challenges

In object detection tasks, occlusion is one of the key challenges that affects the precision and robustness of detection, particularly in scenarios, such as maritime surveillance [1], autonomous shipping [2], and remote sensing [21]. In complex maritime environments, ships frequently overlap due to dense formations, navigation, or docking, causing severe occlusions, compounded by waves, lighting variations, small distant targets, and complex backgrounds. Under these conditions, traditional object detection methods often struggle to identify and locate partially occluded targets effectively, leading to a decrease in detection accuracy.

Occlusion can typically be categorized into partial occlusion, severe occlusion, and dense occlusion. Partial occlusion refers to situations where only key feature areas of the target are partially obscured, allowing for recognition through prominent features. Severe occlusion occurs when most of

the target is hidden, with only a small portion of features exposed, significantly increasing difficulty of detection. Dense occlusion primarily arises between multiple targets, where they overlap, blurring boundary features, and making it difficult for detection models to distinguish individual targets. In many complex maritime scenarios, these types of occlusion are common, making it challenging for ship detection models to accurately recognize and classify targets.

Occlusion causes feature blending and background interference, leading to misclassification, especially in dense scenarios [22]. This issue is particularly challenging in maritime monitoring tasks, where background elements, such as waves, docks, buoys, and coastal buildings can resemble the appearance of ships, further increasing the model's false detection rate. Additionally, occlusion can lead to instability in detection, where the model might produce vastly different predictions for the same target under varying viewpoints or lighting conditions, impacting consistency.

Furthermore, occlusion severely affects the accuracy of detection boxes. Object detection methods typically rely on feature extraction networks to generate object boundaries, but when a target is occluded, its complete shape information is difficult to extract, leading to increased regression errors in bounding boxes. Currently, most ship detection datasets use horizontal bounding boxes (HBBs) for target annotation, including SSDD [23], ShipRSImageNet [24], and SeaShips [25]. In occlusion scenarios, HBB often covers multiple targets, resulting in false detections, and in cases of severe target overlap, both the size and localization accuracy of the bounding boxes are compromised. In contrast, our comprehensive experiments leverage the MID [26], Seaship [25], and SSDD [23] datasets, with MID's oriented bounding box (OBB) annotation providing superior precision for densely occluded maritime targets.

B. Occlusion Detection Method

In the field of ship target detection, occlusion degrades performance by causing feature loss and boundary blurring. In response to the challenge of detecting occluded targets, researchers have proposed various improvement methods in recent years, primarily, including feature enhancement-based methods, attention mechanism-based methods, and feature completion-based methods.

Feature enhancement methods use multiscale fusion to optimize feature extraction, improving robustness for occluded targets. For instance, BiFPN [13] facilitates bidirectional feature flow, allowing higher level features to complement lower-level features, thereby improving the detector's ability to perceive partially occluded targets. Additionally, YOLOv8 employs an improved CSPNet structure to better preserve target boundary information, coupled with dynamic receptive field optimization to more accurately extract features of occluded targets. Recent advances include RSDS [27], which employs a specialized loss function to enhance dense small object detection in remote sensing images by optimizing boundary regression and classification under occlusion; a lightweight multiscale network [28], which improves synthetic

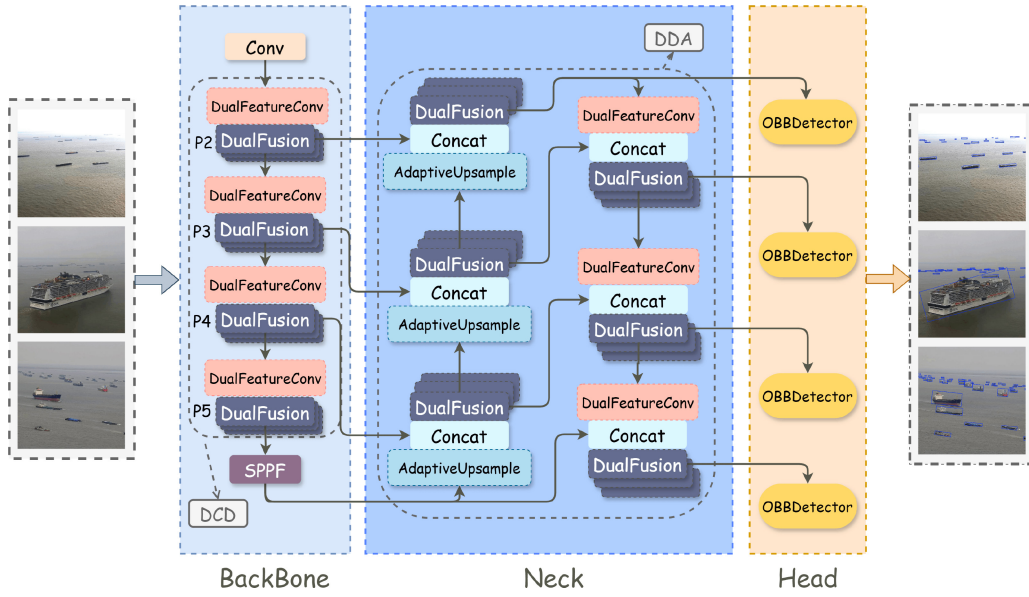


Fig. 1. Overall flow of DSONet. The framework comprises three key components: a Backbone that extracts multiscale feature maps, including P2, P3, P4, and P5, while incorporating DFC and DF module; a Neck module that employs DFC and AU for efficient feature fusion and upsampling; and a Head module responsible for generating precise bounding box predictions.

aperture radar (SAR) ship detection in dense scenes through efficient feature fusion; and ESarDet [29], which leverages contextual information and large effective receptive fields to detect small, occluded ships in complex SAR backgrounds.

Attention mechanisms adaptively weight discriminative features, enhancing detection in complex scenarios. For example, CBAM [30] combines channel attention and spatial attention, allowing the detector to accurately identify the prominent feature regions of the occluded targets. CoAtNet [31], which combines convolutional networks with Transformer structures, enhances the model's ability to recognize highly occluded targets by integrating local perception with global information modeling. Additionally, the Swin Transformer [32] optimizes self-attention computation through a sliding window mechanism, enabling it to more efficiently capture long-range dependency information. However, in ship detection tasks, factors, such as water surface reflection and ship color similarity may further exacerbate the occlusion problem, making it difficult for attention mechanisms alone to fully recover the complete information of occluded targets.

Feature completion methods aim to restore the complete features of occluded targets using contextual information, thereby improving detection accuracy. For example, Amodal Instance Segmentation [33] infers the full shape of a target, enabling the model to predict the contours of occluded areas, thus enhancing the completeness of object detection. Furthermore, DiffusionDet [34] uses a diffusion model to generate complete target features, effectively recovering structural information from the target even in cases of severe occlusion. Existing research indicates that feature completion methods based on generative models exhibit strong generalization ability in object detection tasks. For example, generative adversarial networks (GANs) [35] have been used to generate plausible feature fillings for occluded areas, improving the robustness of detection models. However, in ship detection tasks, due

to significant background interference, generative model-based methods may introduce spurious features, potentially affecting detection accuracy.

In summary, ship occlusion detection continues to face challenges, such as high-density target distribution, long-range small target recognition, and real-time processing requirements. Existing methods still have limitations in complex maritime environments.

III. PROPOSED METHOD

A. Overall Architecture

This article presents a lightweight framework, DSONet, designed for multiscale dense ship occlusion detection. The structure consists of the backbone (DCD), neck (DDA), and detection head, as shown in Fig. 1. The framework realizes multiscale, dense occlusion, and efficient detection by integrating an efficient feature fusion network DF-Det, an upsampling module for deeper feature extraction module-AU, and a rotated target detection head (OBBDetector) matching with the forward feature extraction part, which enhances the accuracy and robustness of ship target detection.

B. DualFeature-Detection (DF-Det)

In maritime target detection tasks, influenced by complex environments and the distribution characteristics of ships, detection models need efficient feature extraction capabilities to address challenges, such as significant variations in target scale, severe dense occlusion, and complex dynamic backgrounds. Inspired by lightweight feature extraction mechanisms like MobileNets [36] and GhostNet [37], this study introduces an improved lightweight feature extraction network, DF-Det.

DF-Det introduces the DFC module for efficient feature extraction and the DF module for feature aggregation. These

modules help reduce computational load, improve detection accuracy, and improve robustness to detect small and occluded targets. Additionally, DF-Det optimizes multiscale representation in target detection tasks through bottom-up, multiscale feature extraction, combined with adaptive feature fusion. The following sections provide detailed descriptions of the DFC and DF modules.

1) *DualFeatureConv (DFC)*: DFC is a key component of the DF-Det architecture. Unlike general-purpose lightweight modules, DFC explicitly balances semantic abstraction and fine-detail preservation by combining pointwise and depthwise convolutions in parallel. This dual-branch design enables enhanced receptive fields while maintaining spatial precision, which is critical for detecting overlapping ship contours in occlusion-heavy scenarios. The resulting module achieves better tradeoffs than purely depthwise or fused convolutions, providing structural novelty tailored to maritime detection. The input of this module is set to $\mathcal{X} \in \mathbb{R}^{B \times C_1 \times H \times W}$, where B , C_1 , H , and W represent the batch size, the number of input channels, and the height and width of the input feature map, respectively.

First, a 1×1 pointwise convolution is applied to $\mathcal{X}$ for channel reduction, decreasing the number of channels in the feature map to $(C_2/2)$. This reduces computational cost while extracting preliminary features

$$\mathcal{Y}_1 = \sigma(\mathcal{X} * \mathcal{W}_1 + \mathbf{b}_1) \in \mathbb{R}^{B \times (C_2/2) \times H \times W} \quad (1)$$

where $\mathcal{W}_1 \in \mathbb{R}^{C_1 \times (C_2/2) \times 1 \times 1}$ represents the 1×1 convolution kernel, $\mathbf{b}_1 \in \mathbb{R}^{(C_2/2)}$ represents the bias term, and $\sigma(\cdot)$ denotes the nonlinear activation function SiLU.

After obtaining the dimension-reduced feature map $\mathcal{Y}_1$, we aim to further extract local spatial information. To achieve this, we introduce a depthwise convolution operation. The core idea of depthwise convolution is to perform spatial convolution independently on each channel, which allows the network to preserve fine-grained information while significantly reducing the number of parameters.

Specifically, we apply a 5×5 depthwise convolutional kernel to $\mathcal{Y}_1$ to better capture spatial context, resulting in a new feature map $\mathcal{Y}_2$

$$\mathcal{Y}_2 = \sigma(\mathcal{Y}_1 \odot \mathcal{W}_2 + \mathbf{b}_2) \in \mathbb{R}^{B \times (C_2/2) \times H \times W} \quad (2)$$

where $\mathcal{W}_2 \in \mathbb{R}^{(C_2/2) \times 1 \times 5 \times 5}$ and $\mathbf{b}_2 \in \mathbb{R}^{(C_2/2)}$, and $\odot$ represents the depthwise convolution operation applied channel-wise.

Finally, the feature maps $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are concatenated along the channel dimension to reconstruct the complete feature representation

$$\mathcal{Y} = \mathbb{C}(\mathcal{Y}_1, \mathcal{Y}_2) \quad (3)$$

where $\mathbb{C}$ denotes channel-wise concatenation and $\mathcal{Y} \in \mathbb{R}^{B \times C_2 \times H \times W}$. This design allows the compressed representation and the spatially enriched features to be jointly preserved, forming a more expressive output. By avoiding a direct 1×1 convolution for channel restoration, potential information loss is mitigated, and the complementary strengths of both branches are retained in a unified feature space.

2) *DualFeatureBottleneck (DFB)*: In object detection tasks, conventional residual block structures (such as ResNet or CSPNet) effectively enhance the flow of information, but they come with a high computational cost. DualFeatureBottleneck (DFB) introduces an efficient bottleneck design that extracts high-level features through two successive DFC operations. It also incorporates depthwise separable convolutions, with residual connections introduced at a stride of two to minimize information loss. To ensure continuity across stages, the input feature map is again denoted as $\mathcal{X}$.

First, initial feature extraction is performed using DFC

$$\mathcal{Y} = \mathcal{DFC}(\mathcal{X}) \in \mathbb{R}^{B \times (C_2/2) \times H \times W}. \quad (4)$$

This step extracts low-level features while compressing the channel dimension, keeping the spatial resolution unchanged.

Next, a 3×3 depthwise convolution layer is applied to further enhance the representation of local features. The output feature map $\mathbb{Y}_1^\dagger$ is defined as

$$\mathbb{Y}_1^\dagger = \sigma(\mathcal{Y} \odot \mathcal{W}_3 + \mathbf{b}_3) \in \mathbb{R}^{B \times (C_2/2) \times H \times W} \quad (5)$$

where $\mathcal{W}_3 \in \mathbb{R}^{(C_2/2) \times 1 \times 3 \times 3}$ denotes a 3×3 convolution kernel, which is applied to $\mathcal{Y}$ to extract local spatial information.

Then, a 1×1 pointwise convolution is applied to restore the channel dimension and enrich the feature representation

$$\mathbb{Y}_2^\dagger = \sigma(\mathbb{Y}_1^\dagger * \mathcal{W}_4 + \mathbf{b}_4) \in \mathbb{R}^{B \times C_2 \times H \times W} \quad (6)$$

where $\mathcal{W}_4 \in \mathbb{R}^{(C_2/2) \times C_2 \times 1 \times 1}$ and $\mathbf{b}_4 \in \mathbb{R}^{C_2}$ are the convolution weights and bias, respectively. This operation completes the channel recovery while embedding higher level semantics into the final feature map.

Subsequently, a residual connection is established by adding the original input $\mathcal{X}$ to the transformed feature map $\mathbb{Y}$. This residual pathway preserves essential information and mitigates gradient vanishing, thereby promoting more stable and efficient training. The details are as follows.

- 1) When the stride $s = 1$, indicating that no spatial downsampling is performed, the input feature map $\mathcal{X}$ is directly fused with the output $\mathbb{Y}_2^\dagger$ via element-wise addition

$$\mathbb{Y} = \mathbb{Y}_2^\dagger + \mathcal{X} \in \mathbb{R}^{B \times C_2 \times H \times W}. \quad (7)$$

In this way, $\mathbb{Y}$ preserves the original spatial resolution and incorporates both low-level contextual cues and high-level semantic features.

- 2) When stride $s = 2$, direct addition becomes infeasible due to mismatched spatial dimensions. To address this, the input feature map $\mathcal{X}$ is downsampled via a stride-2 convolution

$$\mathbb{Y} = \mathbb{Y}_2^\dagger + \mathcal{D}(\mathcal{X}) \in \mathbb{R}^{B \times C_2 \times H \times W} \quad (8)$$

where $\mathcal{D}(\cdot)$ denotes the downsampling operation implemented as

$$\mathcal{D}(\mathcal{X}) = \mathcal{X}_d = \mathcal{X} \odot \mathcal{W}_d \quad (9)$$

with $\mathcal{W}_d \in \mathbb{R}^{C_1 \times 1 \times 3 \times 3}$ being a 3×3 convolution kernel. This operation ensures dimensional alignment while enhancing representational efficiency and reducing computational overhead.

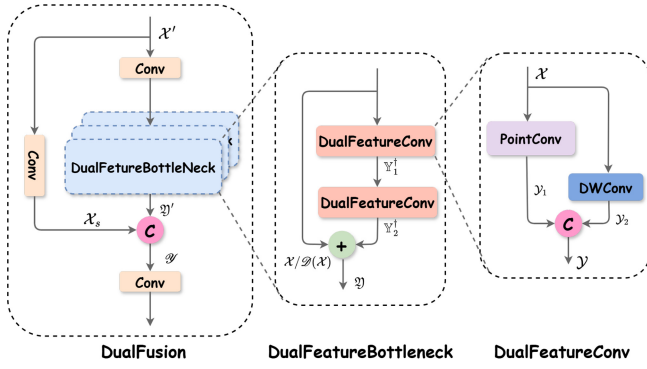


Fig. 2. Structure of DF.

3) *DualFusion (DF)*: DF module is a lightweight feature extraction and fusion unit built upon multiple DFB blocks. The general architecture of DF is illustrated in Fig. 2. It processes the input feature map through a dual-path structure: a main branch for semantic feature extraction and a residual branch for preserving low-level information, thereby enhancing both representational capacity and training stability.

Given input $\mathcal{X}' \in \mathbb{R}^{B' \times C_1' \times H' \times W'}$, the main branch computes high-level features via

$$\mathcal{Y}' = \mathcal{DFB}(\mathcal{X}') \in \mathbb{R}^{B \times C_2 \times H' \times W'} \quad (10)$$

where $\mathcal{DFB}$ serves as the core computational unit, consists of multiple lightweight feature extraction modules (DFC and depthwise separable convolution), which effectively reduce computational complexity while enhancing feature representation capability.

When stride $s = 1$, the dimensions of the feature map $H'' = H'$ and $W'' = W'$, meaning that the size of the feature map remains unchanged. When stride $s = 2$, the dimensions become $H'' = (H'/2)$ and $W'' = (W'/2)$, effectively halving the size of the feature map.

The primary function of the residual connection path is to mitigate information loss and provide a direct gradient flow, improving stability of training. The computation process is as follows.

- 1) When the stride $s = 1$, an identity mapping is used, and the input features are directly preserved

$$\mathcal{X}_s = \mathcal{X}'. \quad (11)$$

This means that the input feature map is passed directly through an identity mapping, preserving the low-level feature information intact. This approach helps prevent gradient vanishing, which is particularly important when training deep networks.

- 2) When the stride $s = 2$, the residual connection performs downsampling on the input feature map using depthwise separable convolution, and adjusts the channel number with a pointwise convolution, ensuring that the output feature map has the same number of channels as the main branch

$$\mathcal{X}_s = \mathcal{DWC}(\mathcal{X}) \quad (12)$$

where $\mathcal{DWC}$ refers to depthwise separable convolution.

Next, a pointwise convolution is applied to match the number of channels

$$\mathcal{X}_s = \mathcal{C}_1(\mathcal{X}_s) \in \mathbb{R}^{B' \times C_2' \times H'' \times W''} \quad (13)$$

where $\mathcal{C}_1$ represents a 1×1 convolution operation used to adjust the channel numbers, ensuring that the output from the residual connection matches the channel dimensions of the main branch output.

After computing the outputs from both the main and residual branches, we perform an element-wise addition to obtain the final output feature map $\mathcal{Y}$

$$\mathcal{Y} = \mathcal{Y}' + \mathcal{X}_s \in \mathbb{R}^{B \times C_2 \times H' \times W'} \quad (14)$$

where $\mathcal{Y}'$ represents the high-level features from the main branch, and $\mathcal{X}_s$ denotes the low-level features from the residual path. This addition facilitates the fusion of hierarchical features, promoting effective gradient flow and enhancing representational capacity.

C. AdaptiveUpsample (AU)

Building upon the DF-Det framework, an AU module is introduced as a lightweight and content-adaptive upsampling component. AU dynamically generates reorganization kernels conditioned on local feature context, enabling more precise spatial refinement. This design enhances detection robustness, particularly under complex scenes with dense object occlusions. Given an input feature map $\mathcal{X} \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H} \times \hat{W}}$, a convolutional layer is first applied to reduce the channel dimension, producing an intermediate representation $\mathcal{Z}_1$

$$\mathcal{Z}_1 = \mathcal{C}(\mathcal{X}) \in \mathbb{R}^{\hat{B} \times C_{\text{mid}} \times \hat{H} \times \hat{W}} \quad (15)$$

where C_{mid} denotes the compressed channel size.

Subsequently, the intermediate feature map $\mathcal{Z}_1$ is encoded via a convolutional layer to generate the upsampling kernels

$$\mathcal{Z}_2 = \mathcal{C}(\mathcal{Z}_1) \in \mathbb{R}^{\hat{B} \times (k_{\uparrow} \cdot \text{scale})^2 \times \hat{H} \times \hat{W}}. \quad (16)$$

Here, $(k_{\uparrow} \cdot \text{scale})^2$ denotes the number of kernel weights produced at each spatial location, which are then reshaped into $k_{\uparrow} \times k_{\uparrow}$ filters to enable content-adaptive spatial reorganization in the subsequent reconstruction step.

After the convolution operation, $\mathcal{Z}_2$ undergoes a PixelShuffle operation for pixel reorganization, which enhances the spatial resolution of the feature map. Through PixelShuffle, the spatial dimensions of the feature map are expanded from $\hat{H} \times \hat{W}$ to $\hat{H}_{\uparrow} \times \hat{W}_{\uparrow}$, resulting in the output feature map $\mathcal{Z}_3$

$$\mathcal{Z}_3 = \mathcal{PSF}(\mathcal{Z}_2) \in \mathbb{R}^{B \times (k_{\uparrow}^2) \times \hat{H}_{\uparrow} \times \hat{W}_{\uparrow}} \quad (17)$$

where $\hat{H}_{\uparrow} = \hat{H} \times \text{scale}$ and $\hat{W}_{\uparrow} = \hat{W} \times \text{scale}$, meaning the spatial dimensions are expanded after upsampling. $k_{\uparrow}^2$ represents the number of channels after the spatial reorganization, which is obtained by rearranging the convolutional channels through PixelShuffle.

Next, a softmax operation is applied to normalize $\mathcal{Z}_3$

$$\mathcal{Z}_3 = \text{softmax}(\mathcal{Z}_3, \text{dim} = 1) \quad (18)$$

which normalizes the features of each pixel along the channel dimension so that the sum of the reconstruction weights for each pixel across all channels equals 1. This assigns weights across channels, enhancing the accuracy of feature map reconstruction.

Meanwhile, the input feature map $\mathcal{X}$ undergoes a nearest-neighbor upsampling operation to upscale its spatial dimensions. This operation expands the spatial size of the input feature map from $\hat{H} \times \hat{W}$ to $\hat{H}_\uparrow \times \hat{W}_\uparrow$

$$\mathcal{X}_\uparrow = \mathcal{U}(\mathcal{X}) \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H}_\uparrow \times \hat{W}_\uparrow} \quad (19)$$

where $\mathcal{U}$ represents the upsampling operation. This operation increases the spatial resolution of the input feature map, providing higher resolution for subsequent reconstruction and weighting operations.

Subsequently, the upsampled feature map $\mathcal{X}_\uparrow$ undergoes the Unfold operation, which utilizes a sliding window mechanism to transform local region feature information into separate blocks, enabling convolution operations

$$\mathcal{X}_{\text{unf}} = \mathcal{UF}(\mathcal{X}_\uparrow) \in \mathbb{R}^{\hat{B} \times (\hat{k}_{\text{up}}^2 \times \hat{C}) \times \hat{H}_{\text{up}} \times \hat{W}_{\text{up}}} \quad (20)$$

where the $\mathcal{UF}$ operation transforms the feature map into smaller blocks with multiple channels, allowing effective feature fusion at each pixel location for subsequent weighting operations.

Finally, feature weighting and reconstruction are performed. The goal is to apply weights to each pixel in both the spatial and channel dimensions, generating the final upsampled feature map. In this formulation, the final upsampled feature map $\mathcal{Y}'$ is obtained by weighting and aggregating the unfolded feature blocks

$$\mathcal{Y}' = \sum_{i=1}^{\hat{C}} \mathcal{Z}_3^{(i)} \otimes \mathcal{X}_{\text{unf}}^{(i)} \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H}_\uparrow \times \hat{W}_\uparrow} \quad (21)$$

where $\mathcal{Z}_3^{(i)}$ is the weight map for channel i , $\mathcal{X}_{\text{unf}}^{(i)}$ the corresponding unfolded feature block, and $\otimes$ denotes element-wise multiplication. This adaptive weighting mechanism emphasizes salient local features during reconstruction, yielding a higher-resolution representation. Here, $\hat{C}$ denotes the number of channels in $\mathcal{Y}'$. The calculation process of the AU module is also presented in Algorithm 1.

D. OBB Detector

In the DSONet framework, we propose an OBB detection head to improve multiscale ship detection under complex maritime conditions. Unlike conventional HBB methods that struggle with occlusion, elongated structures, and arbitrary orientations, the OBB head introduces an additional angle regression branch, enabling more accurate and rotation-aware localization of ships.

To accommodate variations in object size and distribution, DSONet adopts a hybrid feature hierarchy that combines feature pyramid networks (FPN) and path aggregation networks (PAN), enabling the prediction of oriented objects across four scales (P2–P5). At the P2 level, the network focuses on tiny, distant ships that are often buried in background clutter and

Algorithm 1 AU Module

Input: Input feature map $\mathcal{X} \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H} \times \hat{W}}$
Output: Output feature map $\mathcal{Y}' \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H}_\uparrow \times \hat{W}_\uparrow}$

- 1: // **Step 1: Channel Compression**
- 2: $\mathcal{Z}_1 \leftarrow \mathcal{C}_1(\mathcal{X}) \in \mathbb{R}^{\hat{B} \times \hat{C}_{\text{mid}} \times \hat{H} \times \hat{W}}$
- 3: // **Step 2: Upsampling Kernel Generation**
- 4: $\mathcal{Z}_2 \leftarrow \mathcal{C}_2(\mathcal{Z}_1) \in \mathbb{R}^{\hat{B} \times (\hat{k}_\uparrow \cdot \text{scale})^2 \times \hat{H} \times \hat{W}}$
- 5: // **Step 3: PixelShuffle and Normalization**
- 6: $\mathcal{Z}_3 \leftarrow \text{softmax}(\mathcal{PSF}(\mathcal{Z}_2), \text{dim} = 1)$
- 7: ▷ PixelShuffle:
- 8: $\mathcal{PSF}(\mathcal{Z}_2) \in \mathbb{R}^{\hat{B} \times \hat{k}_\uparrow^2 \times \hat{H}_\uparrow \times \hat{W}_\uparrow}$,
- 9: $\Rightarrow \mathcal{Z}_3[b, c, r \cdot h + i, r \cdot w + j] \leftarrow \mathcal{Z}_2[b, c \cdot r^2 + i \cdot r + j, h, w]$
- 10: for $i, j \in \{0, \dots, r-1\}$
- 11: where $\hat{H}_\uparrow = \hat{H} \cdot \text{scale}$, $\hat{W}_\uparrow = \hat{W} \cdot \text{scale}$
- 12: // **Step 4: Nearest Neighbor Upsampling**
- 13: $\mathcal{X}_\uparrow \leftarrow \mathcal{U}(\mathcal{X})$
- 14: ▷ Nearest-neighbor upsampling:
- 15: $\mathcal{X}_\uparrow \in \mathbb{R}^{\hat{B} \times \hat{C} \times \hat{H}_\uparrow \times \hat{W}_\uparrow}$,
- 16: $\Rightarrow \mathcal{X}_\uparrow[b, c, r \cdot h + i, r \cdot w + j] \leftarrow \mathcal{X}[b, c, h, w]$
- 17: for $i, j \in \{0, \dots, r-1\}$
- 18: with $\hat{H}_\uparrow = \hat{H} \cdot \text{scale}$, $\hat{W}_\uparrow = \hat{W} \cdot \text{scale}$
- 19: // **Step 5: Unfold Operation for Local Region Encoding**
- 20: $\mathcal{X}_{\text{unf}} \leftarrow \mathcal{UF}(\mathcal{X}_\uparrow)$
- 21: ▷ $\mathcal{X}_{\text{unf}} \in \mathbb{R}^{\hat{B} \times (\hat{k}_\uparrow^2 \cdot \hat{C}) \times \hat{H}_\uparrow \times \hat{W}_\uparrow}$
- 22: // **Step 6: Adaptive Feature Weighting and Aggregation**
- 23: $\mathcal{Y}' \leftarrow \sum_{i=1}^{\hat{C}} \mathcal{Z}_3^{(i)} \otimes \mathcal{X}_{\text{unf}}^{(i)}$
- 24: **return** $\mathcal{Y}'$

TABLE I
OCCLUSION INSTANCE STATISTICS

Instances	Illustration	Number
Unobstructed Instances	$\leq 10\%$	114090
Slightly Obstructed Instances	10%-20%	9287
Partially Obstructed Instances	20%-50%	10036
Severely Obstructed Instances	50%-90%	1634
Completely Obstructed Instances	$\geq 90\%$	837
Total Instances	-	135884

prone to feature degradation. To address this, the AU module is applied to enhance spatial resolution, while Focal Loss is used to emphasize hard samples, improving small-object detection robustness.

As scale increases, the network adapts to more complex structures and interactions. At P3, small ships in congested areas benefit from OBB's alignment with object orientation, reducing ambiguity caused by dense placement. P4 targets midsize vessels with irregular shapes and occlusion, where DF-Det's semantic representations further refine contour localization. At P5, which deals with large ships, such as carriers or freighters, conventional HBB often result in misalignment and background redundancy. In contrast, OBBs achieve tighter and more semantically consistent localization by regressing rotation angles, enhancing overall detection precision.

IV. EXPERIMENT

A. Datasets

Publicly available datasets for occluded and overlapping ship detection remain scarce. To bridge this gap, we employ

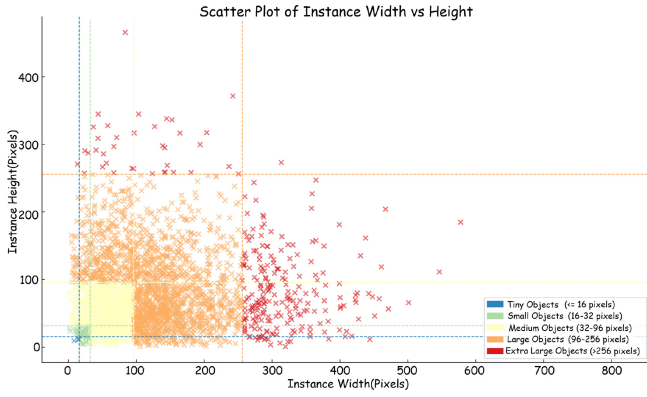


Fig. 3. Scatter plot of instance width versus height in MID.

TABLE II
SPECIFIC EXPERIMENTAL DETAILS

Dataset	Epochs	Optimizer	Batch	Lr0	Momentum	Weight_decay
MID	100	AdamW	16	0.002	0.9	5e-4
SeaShip	100	AdamW	16	0.002	0.9	5e-4
SSDD	300	AdamW	16	0.002	0.9	5e-4

the MID dataset [26], which comprises 5673 images and 135884 finely annotated instances using oriented bounding boxes, sourced from 43 distinct navigable water areas. Captured in diverse scenarios: varying weather, docking maneuvers, small target clusters, and partial occlusions. MID offers a rigorous benchmark for complex maritime detection tasks. Fig. 3 visualizes the distribution of instance widths versus heights, and Table I details occlusion types along with their sample counts.

Furthermore, to enhance the model's capability in detecting small objects in high-resolution optical remote sensing imagery, we employed the Seaship dataset [25], which comprises approximately 7000 images annotated with HBBs tailored for small object detection. The dataset includes fine-grained categories, such as cargo ships, oil tankers, container ships, and warships, with annotations addressing complex scenarios, such as occlusion and partial visibility. It presents challenging conditions, including densely packed small targets and interference from complex sea states (e.g., wave occlusion and cloud coverage).

In addition, to assess the generalization ability of the proposed method in multimodal remote sensing contexts, we further utilized the SSDD dataset [23] to explore its robustness on SAR imagery.

B. Training Details

Our experiments were conducted on a system running Ubuntu 20.04, equipped with an Intel Xeon Platinum 8369B CPU 2.90 GHz and Nvidia V100 32 GB GPUs. Detailed experimental settings are provided in Table II.

C. Evaluation Metrics

To comprehensively assess detection performance, we adopt multiple standard metrics. Specifically, we report mean Average Precision (mAP) at three Intersection over Union

TABLE III
ABLATION EXPERIMENTS ON MID

DF-Det:	AU:	mAP _{.50} (%)	mAP _{.75} (%)	mAP _{.50:.95} (%)	Params(M)	FLOPs(G)
×	×	91.9	79.7	71.6	3.08	8.4
✓	×	95.0	83.5	74.0	1.61	8.8
×	✓	92.3	80.6	72.1	3.15	8.5
✓	✓	95.2	84.4	75.0	1.81	9.8

(IoU) thresholds: mAP_{.50} for loose matching, mAP_{.75} for stricter localization, and mAP_{.50:.95}, which averages results over ten IoU thresholds from 0.50 to 0.95 with a step size of 0.05

$$\text{mAP}_{.50:.95} = \frac{1}{N} \sum_{i=1}^N \text{AP}_{\text{IoU}_i}, \quad N = 10. \quad (22)$$

In addition to mAP, we include Precision (P) and Recall (R) to further assess precision and completeness of the model classification. To assess computational efficiency, we also report the number of parameters (Params) and floating point operations (FLOPs). Furthermore, considering the multiscale nature of the ship detection task, we report AP_s, AP_m, and AP_l, which measure the detection performance for small (< 32² pixels), medium (32² – 96² pixels), and large targets (> 96² pixels) respectively. These scale-aware metrics provide additional insights into the model's capability across object sizes. Together, these metrics provide a balanced evaluation of both detection accuracy and inference complexity.

D. Ablation Experiments

To demonstrate the superiority of the DSONet network, we conducted ablation experiments on the MID dataset, using mAP_{.50}, mAP_{.50:.95}, and Params as quantitative metrics. Accordingly, Table III presents the impact of each component.

1) *Effect of the DF-Det Structure*: Compared to the baseline, the DF-Det structure improves mAP_{.50} by 3.1%, and mAP_{.50:.95} by 2.4%, while reducing the number of parameters by 47.7% and increasing FLOPs by 4.8%. The DF-Det framework modifies the baseline YOLOv8n-obb by adjusting the network depth, adding an extra P2 layer for small object detection, replacing standard convolutions with DFC, and utilizing the DF structure for enhanced feature fusion. The results show that the DF-Det framework can significantly improve detection accuracy while reducing model parameters by nearly 50%.

2) *Effect of the AU Module*: Compared to the baseline, the use of the AU module improves mAP_{.50} by 0.4% and mAP_{.50:.95} by 0.5%, while increasing the number of parameters by 2.3% and FLOPs by 1.2%. The results indicate that the AU module, enhances multiscale feature fusion. Although there is a slight increase in both the parameters and FLOPs, it still contributes to the improvement in the detection accuracy.

3) *Effect of DSONet*: The DSONet model, which combines the DF-Det framework and the AU module, demonstrates significant superiority in both mAP_{.50} and mAP_{.50:.95}. Compared to baseline, our network improves mAP_{.50} by 3.3% and mAP_{.50:.95} by 3.4%, while reducing the number of parameters by 41.2% and increasing FLOPs by 16.7%. DSONet has

TABLE IV
COMPARISON EXPERIMENTS ON MID

Method	P	R	mAP _{.50} (%)	mAP _{.75} (%)	mAP _{.50:.95} (%)	AP _s (%)	AP _m (%)	AP _l (%)	Params(M)	FLOPs(G)
YOLOv3t-obb [38]	0.927	0.532	71.4	60.2	53.8	38.99	82.28	90.28	12.65	19.9
YOLOv5n-obb	0.942	0.82	91.1	78.2	70.1	72.29	91.47	94.10	2.58	7.4
YOLOv6n-obb [39]	0.932	0.811	90.5	74.3	67.0	67.24	89.84	93.44	4.25	11.8
YOLOv8n-obb	0.945	0.83	91.9	79.7	71.6	73.83	92.66	94.56	3.08	8.4
YOLOv9t-obb [11]	0.934	0.822	91.1	78.1	69.8	72.08	89.75	94.64	2.05	8.1
YOLOv10n-obb [12]	0.944	0.831	92.1	80.3	71.9	74.17	90.72	94.23	2.67	8.2
YOLOv11n-obb	0.938	0.821	90.9	78.2	70.0	72.28	90.43	94.87	2.66	6.7
YOLOv12n-obb [40]	0.935	0.82	90.8	77.6	69.2	70.67	90.09	95.17	2.59	6.1
BiFPN (v8n based) [13]	0.941	0.831	91.6	79.8	71.4	73.14	91.43	94.23	2.59	6.1
ODConv (v8n based) [41]	0.941	0.823	91.4	78.6	70.3	71.19	89.54	94.21	3.14	6.0
CBAM (v8n based) [30]	0.940	0.831	92.2	79.6	71.4	73.10	92.36	93.89	3.22	9.2
MobileNetV4 (v10n based) [42]	0.927	0.830	91.2	77.6	69.2	70.87	91.45	93.87	5.68	22.7
StarNet (v10n based) [43]	0.934	0.832	91.9	78.1	69.3	68.19	89.44	93.80	2.19	6.7
Ours	0.96	0.92	95.2	84.4	75.0	80.34	92.38	95.43	1.81	9.8

TABLE V
COMPARISON EXPERIMENTS ON SEASHIP

Method	P	R	mAP _{.50} (%)	mAP _{.75} (%)	mAP _{.50:.95} (%)	AP _s (%)	AP _m (%)	AP _l (%)	Params(M)	FLOPs(G)
YOLOv3t [38]	0.888	0.883	92.7	73.4	62.9	56.89	82.99	79.91	12.13	18.9
YOLOv5n	0.924	0.879	94.6	79.1	67.3	56.28	87.60	81.58	2.50	7.1
YOLOv6n [39]	0.919	0.864	93.6	79.9	67.2	55.10	86.93	82.87	4.23	11.8
YOLOv8n	0.919	0.886	94.4	79.1	66.8	54.07	86.90	82.32	3.01	8.1
YOLOv9t [11]	0.913	0.864	91.9	70.3	62.1	51.52	77.07	73.01	1.97	7.8
YOLOv10n [12]	0.830	0.792	88.0	75.4	63.4	48.08	82.12	81.71	2.27	8.4
YOLOv11n	0.918	0.882	94.0	80.1	67.0	49.26	84.98	81.21	2.58	6.3
YOLOv12n [40]	0.910	0.877	93.9	79.8	66.6	48.39	83.23	80.61	2.56	6.0
BiFPN (v8n based) [13]	0.938	0.882	94.6	80.8	67.8	51.12	87.56	82.70	1.99	7.1
ODConv (v8n based) [41]	0.921	0.870	93.4	77.6	65.6	49.72	86.52	81.63	3.07	5.7
CBAM (v8n based) [30]	0.934	0.887	94.5	80.9	68.5	49.72	86.54	81.64	3.15	9.1
MobileNetV4 (v10n based) [42]	0.820	0.781	86.5	70.9	61.2	40.57	78.95	79.73	5.28	21.2
StarNet (v10n based) [43]	0.853	0.804	89.5	74.5	63.8	39.48	82.63	80.48	2.19	6.7
Ours	0.933	0.889	94.9	81.8	68.9	55.92	88.37	82.51	1.80	9.7

been optimized for multiscale targets, dense occlusion scenarios, low computational resource environments, and complex weather conditions, and its efficiency has been experimentally validated on the MID dataset. And the effectiveness of our multiscale design is further illustrated in Fig. 4. Compared to YOLOv8n, DSONet achieves higher detection accuracy, lower computational cost, and is applicable to a wider range of scenarios, making it suitable for intelligent maritime surveillance, drone patrol detection, autonomous navigation obstacle avoidance, and other related applications.

E. Comparison Experiments

We selected several representative object detectors, including the single-stage detectors YOLOv3t [38], YOLOv5n, YOLOv6n [39], YOLOv8n, YOLOv9n [11], YOLOv10n [12], YOLOv11n, and YOLOv12n [40], as well as the existing multiscale and attention-based approaches BiFPN [13], ODConv [41], CBAM [30], MobileNetV4 [42], and StarNet [43]. The comparison results are presented in Table IV.

The experimental results show that DSONet outperforms other models in several key performance metrics, demonstrating exceptional detection capability and computational efficiency. In terms of accuracy, DSONet achieves 95.2%

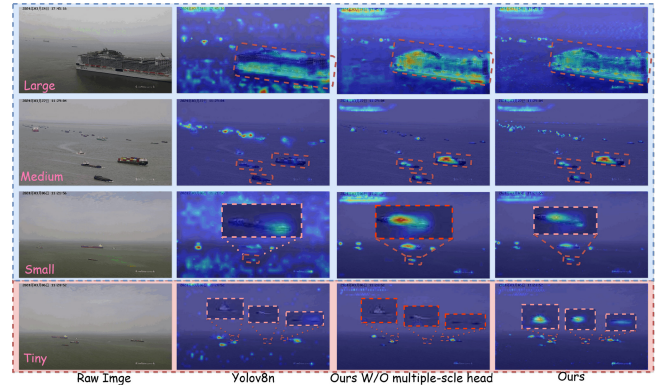


Fig. 4. Heatmaps of different models. From left to right: YOLOv8n, our model without multiscale heads, and our full model. The proposed approach produces more focused and accurate responses, especially for small and densely packed ships.

mAP_{.50} and 84.4% mAP_{.75}, which is a significant improvement, especially in multiscale and densely occluded scenarios, where its detection capability is more stable. For the mAP_{.50:.95} metric, DSONet reaches 75.0%, surpassing other models and demonstrating its strong generalization ability at different IoU thresholds, enabling more accurate detection of ship targets in complex backgrounds.

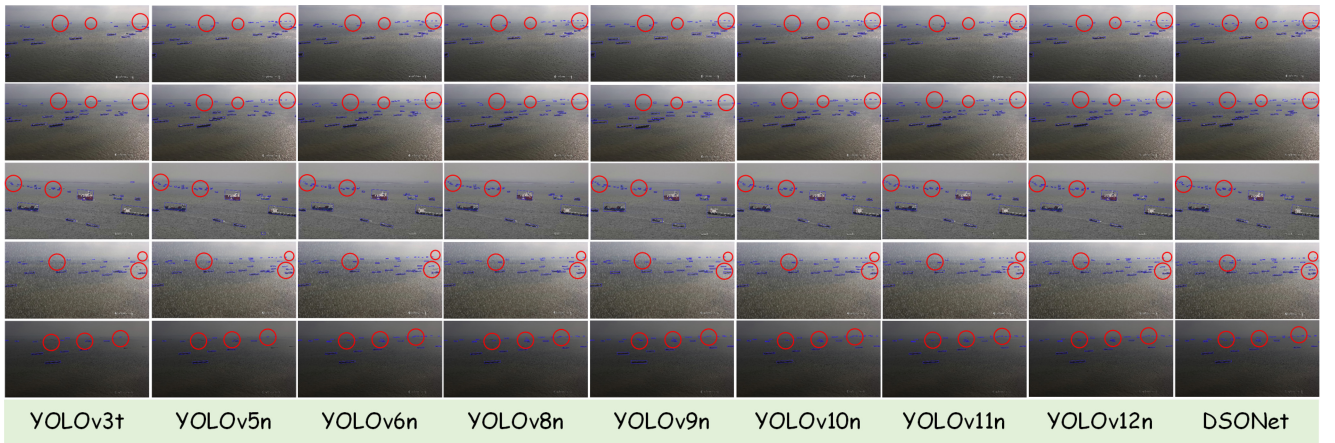


Fig. 5. Visualization of dense occlusion of small object detection results of different models on the MID. Input images with small objects are shown, along with detection results from various models and the proposed DSONet model, each marked with bounding boxes. Additionally, red circles are used to specifically denote instances where the models successfully identified or missed small objects under dense occlusion conditions.



Fig. 6. Visualization of medium and large object detection results of different models on the MID. Input images with medium and large objects are shown, along with detection results from various models and the proposed DSONet model, each marked with bounding boxes. Additionally, red circles are used to specifically denote instances where the models successfully identified or missed objects under dense occlusion conditions.

To further assess the effectiveness of DSONet in multiscale detection tasks, we additionally report average precision for small (AP_s), medium (AP_m), and large (AP_l) objects. DSONet achieves 80.34% AP_s , 92.38% AP_m , and 95.43% AP_l , all of which outperform the compared models. The substantial gain in AP_s confirms DSONet's strong capacity to detect small, low-resolution ships that are often missed by conventional detectors, while improvements in AP_m and AP_l verify its robustness in handling medium and large vessels with varying shapes and occlusion patterns.

We further evaluated the robustness and generalization ability of the proposed method by conducting cross-dataset experiments on two publicly available ship detection datasets: SeaShips and SSDD. As shown in Table V, DSONet achieves 94.9% mAP_{50} , 81.8% mAP_{75} , and 68.9% $mAP_{50:95}$ on the SeaShips dataset, along with 52.92% AP_s , 88.37% AP_m , and 82.51% AP_l , clearly outperforming all baseline models. This demonstrates the model's superior capability in handling small, distant ships and complex sea-state interference. Meanwhile, as shown in Table VI, DSONet achieves 98.7% mAP_{50} and 73.2% $mAP_{50:95}$ on SSDD, outperforming SAR-oriented detectors, such as FEPS-Net and RSNet despite the significant modality difference.

In terms of computational efficiency, DSONet has a parameter count of only 1.81 M, significantly lower than other models, indicating that its architecture dramatically reduces storage requirements while maintaining high accuracy. Additionally, DSONet performs well in terms of FLOPs, reaching 9.8G.

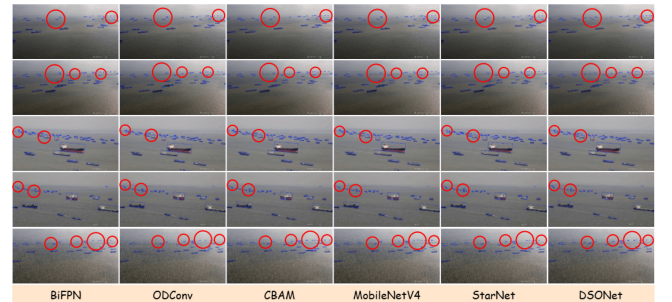


Fig. 7. Visualization of detection results for DSONet and existing methods (BiFPN, ODConv, CBAM, MobileNetV4, StarNet) on the MID, with bounding boxes. Red circles indicate instances of successful or missed target detection under dense occlusion conditions.

While this is slightly higher than other models, the substantial improvement in detection accuracy demonstrates that DSONet achieves the optimal balance between computational efficiency and detection performance, enabling it to run efficiently even in resource-constrained environments.

To provide a more intuitive understanding of the detection performance of different models in multiscale targets, densely occluded objects, and complex background scenarios, we further compared the actual detection results of these models on the MID dataset. To thoroughly demonstrate the superiority of the proposed DSONet in multiscale dense ship occlusion detection scenarios, the detection results of the models in Table IV on the same input images are shown in Fig. 5. In the figure, the blue boxes indicate the targets detected by the

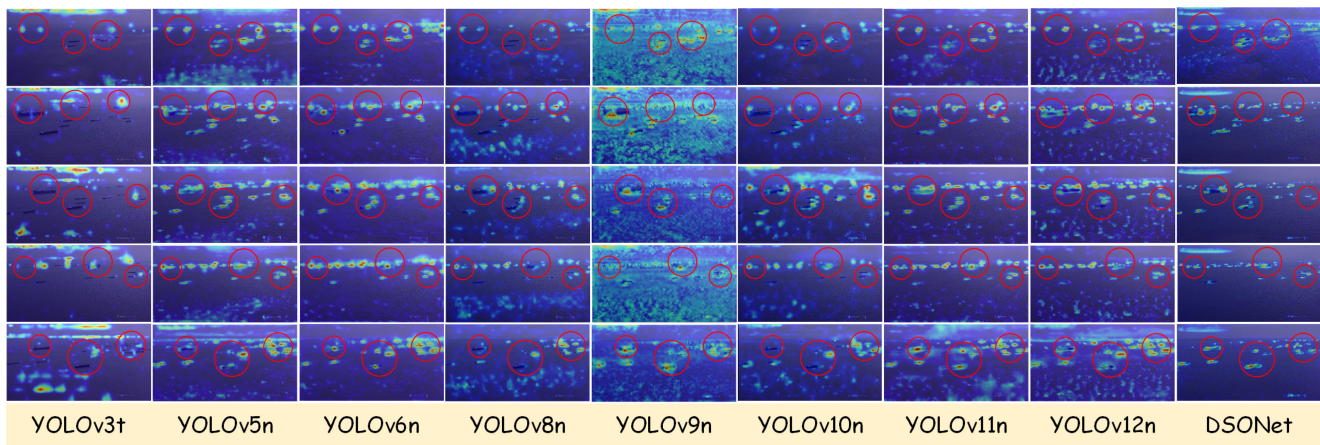


Fig. 8. Heatmaps of dense occlusion of small object detection across different models on the MID. This figure presents heatmaps showing the dense occlusion handling in small object detection for various models and the proposed DSONet model. Each heatmap highlights areas of interest with red circles, indicating regions where the models successfully identified or missed small objects under dense occlusion conditions.

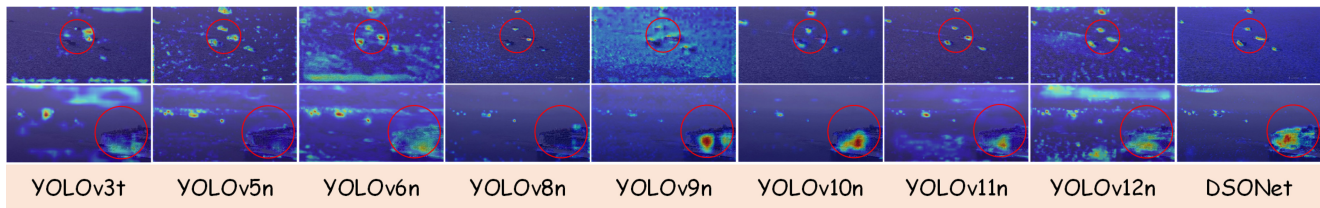


Fig. 9. Heatmaps of medium and large object detection across different models on the MID. This figure presents heatmaps showing the dense occlusion handling in medium and large object detection for various models and the proposed DSONet model. Each heatmap highlights areas of interest with red circles, indicating regions where the models successfully identified or missed medium and large objects.

models, while the red circles highlight the key comparison areas. The detection results clearly demonstrate that DSONet outperforms other detectors in dense occlusion scenarios, particularly in cases of target overlap, heavy occlusion, and small targets at long distances. In the first three rows of Fig. 5, all models are able to identify relatively clear ship targets. However, in the detection of small distant targets, models, such as YOLOv3t, YOLOv5n, YOLOv6n, and YOLOv8n, due to limited feature extraction capabilities, fail to correctly detect some targets. In contrast, DSONet, with its P2-level feature introduction and the feature enhancement of AU, can accurately detect small targets at long distances, effectively preventing missed detections.

In the fourth and fifth rows of Fig. 5, the images show the performance of different detectors in complex backgrounds. Regarding false detection control, models, such as YOLOv3t, YOLOv6n, and YOLOv9n are prone to false detections in areas with strong interference factors, such as waves and shadows. YOLOv8n, YOLOv10n, YOLOv11n, and YOLOv12n have fewer false detections but still produce spurious bounding boxes in complex backgrounds. On the other hand, DSONet is more accurate in distinguishing between target and background, significantly reducing the false detection rate. Furthermore, in terms of stability of bounding box and precision of target boundary, YOLOv3t's bounding boxes tend to drift or expand over time, while models, such as YOLOv9n and YOLOv10n still face target merging issues in occlusion scenarios. DSONet, with its strengthened multiscale

feature fusion, ensures clearer target boundaries. Even in low-contrast, long-distance, or complex background conditions, the detection boxes remain precise without misalignment, target merging, or loss.

In addition, Fig. 6 presents a comparison of the visualization results for medium and large object detection tasks across different models mentioned in Fig. 5. DSONet demonstrates a clear advantage in detecting medium and large objects, with superior accuracy in identifying and localizing targets. Its performance in complex scenarios exceeds that of other YOLO models, demonstrating stronger stability and precision.

To evaluate DSONet's performance in dense ship occlusion, Fig. 7 compares bounding box outcomes on the MID dataset for DSONet and existing methods, including BiFPN, ODConv, CBAM, MobileNetV4, and StarNet. The figure displays overlapping ships with model bounding boxes, and red circles mark areas of varying detection, especially for small and overlapped targets, revealing insights into scale and occlusion challenges. Fig. 7 shows BiFPN, despite being relatively effective, struggles with inconsistent bounding boxes in occluded areas and misses small targets, while ODConv, CBAM, MobileNetV4, and StarNet perform worse. DSONet provides accurate bounding boxes, especially in red-circled areas, excelling in multiscale and occluded ship detection.

Furthermore, to analyze target attention regions, feature extraction capabilities, and sensitivity to occluded targets in different detection models, we use heatmaps to compare the feature responses of each model. Figs. 8 and 9 present the

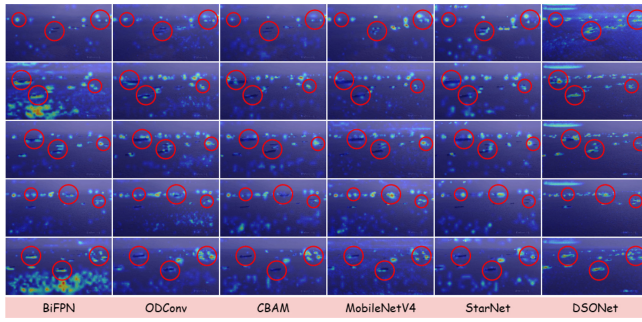


Fig. 10. Heatmaps of object detection for DSONet and existing methods (BiFPN, ODConv, CBAM, MobileNetV4, StarNet) on the MID. Heatmaps illustrate dense occlusion handling, with red circles indicating regions of successful or missed detection.

heatmaps of the response to the characteristics of the compared models and our models on the MID dataset, with red circles highlighting the key comparison areas.

Comparative heatmap analysis reveals that YOLOv3t struggles with detecting small, distant ships. In YOLOv8n, YOLOv9n, and YOLOv10n, ship boundaries are blurred, with merging and overflow in crowded regions. YOLOv11n and YOLOv12n show improved accuracy, but some targets remain missed. Notably, DSONet provides sharper delineation of ship contours, with significantly enhanced activations, especially in red-circled areas. Models like YOLOv5n, YOLOv6n, YOLOv8n, and YOLOv9n exhibit weak responses for distant or small ships, often resulting in missed detections. While YOLOv10n, YOLOv11n, and YOLOv12n show moderate improvement, they still suffer from information loss for distant objects. In contrast, DSONet, supported by its AU module, consistently provides clear and strong responses even for small, distant targets.

Additionally, DSONet shows a clear advantage in detecting medium and large targets. Models like YOLOv3t, YOLOv5n, and YOLOv6n exhibit weak and scattered activation responses, leading to higher missed rates. YOLOv8n through YOLOv12n show moderate improvement, but their heatmaps remain diffuse under complex conditions, hindering precise localization. In contrast, DSONet consistently produces sharper, more concentrated activations, accurately delineating medium and large objects while minimizing both false negatives and positives. This improvement is attributed to the combined effect of the optimized DF-Det architecture and the AU module, which improve spatial focus and recognition accuracy.

Fig. 10 compares heatmap responses of DSONet and baseline methods from Tables IV and V on the MID dataset, evaluating feature extraction and sensitivity to occluded targets. The heatmaps depict activation patterns under dense occlusion, with red circles highlighting regions of varying response strength, offering insights into the models' handling of background interference and occluded ship features in maritime settings. Analysis indicates that BiFPN, while relatively effective among existing methods, shows diffuse activation in occluded areas, missing small ships, while ODConv, CBAM, MobileNetV4, and StarNet exhibiting weaker responses.

TABLE VI
COMPARISON EXPERIMENTS ON SSDD

Method	mAP _{.50} (%)	mAP _{.50:.95} (%)	Params(M)	FLOPs(G)
YOLOv3t [38]	96.0	71.5	12.13	18.9
YOLOv5n	97.9	72.6	2.50	7.1
YOLOv6n [39]	96.9	71.2	4.23	11.8
YOLOv8n	98.2	72.6	3.01	8.1
YOLOv9t [11]	98.2	71.2	2.01	7.8
YOLOv10n [12]	97.0	71.3	2.71	8.4
YOLOv11n	98.0	72.0	2.59	6.4
YOLOv12n [40]	97.3	70.1	2.52	6.0
YOLOv13n [44]	97.3	69.8	2.49	6.2
BiFPN [13]	98.4	72.5	1.99	7.1
ODConv [41]	98.2	72.9	3.07	5.7
CBAM [30]	98.4	73.0	3.15	9.1
MobileNetV4 [42]	96.6	69.4	5.28	21.2
StarNet [43]	97.0	70.2	2.19	6.7
FEPS-Net [45]*	96.0	73.1	37.31	-
BL-Net [46]*	95.25	-	47.81	-
CS ⁿ Net [47]*	97.1	64.9	12.2	21.7
RSNet [48]*	98.4	72.5	1.49	5.1
Ours	98.7	73.2	1.80	9.7

* Denotes data obtained from their papers

However, DSONet displays focused activation, notably in red-circled regions, where its enhanced feature recovery improves ship contour delineation under complex backgrounds. This underscores the models' differing abilities in addressing occlusion and interference, with DSONet performing superiorly.

As shown in Fig. 11, and based on the evaluation criteria in Table I, the detection performance of different models is presented as heatmaps at varying occlusion levels. Among all models, DSONet consistently outperforms the others, especially under medium and high occlusion levels, where traditional detectors typically fail. The heatmap values for DSONet are significantly higher and more uniformly distributed, demonstrating its robustness against visual interference and its ability to effectively recover discriminative features from partially occluded targets. These results highlight the capability of DSONet to address the challenges of dense, multiscale occlusions, validating the efficacy of its lightweight backbone and adaptive upsampling strategy in improving detection accuracy and generalization in complex maritime environments.

In general, DSONet outperforms other models in multiscale detection, target separation, and background suppression, making it more suitable for ship detection tasks in complex maritime scenarios.

V. DISCUSSIONS

The proposed DSONet framework significantly enhances ship detection accuracy, particularly under multiscale and dense occlusion conditions. Experimental results show that DSONet outperforms lightweight baselines, such as YOLO series models and other competitive models, achieving higher accuracy with lower parameter count and faster inference. This is largely attributed to the DF-Det backbone, where the DFC and DF modules improve feature representation while reducing redundant computation, enabling efficient deployment on edge devices. Moreover, the AU module enhances high-low feature

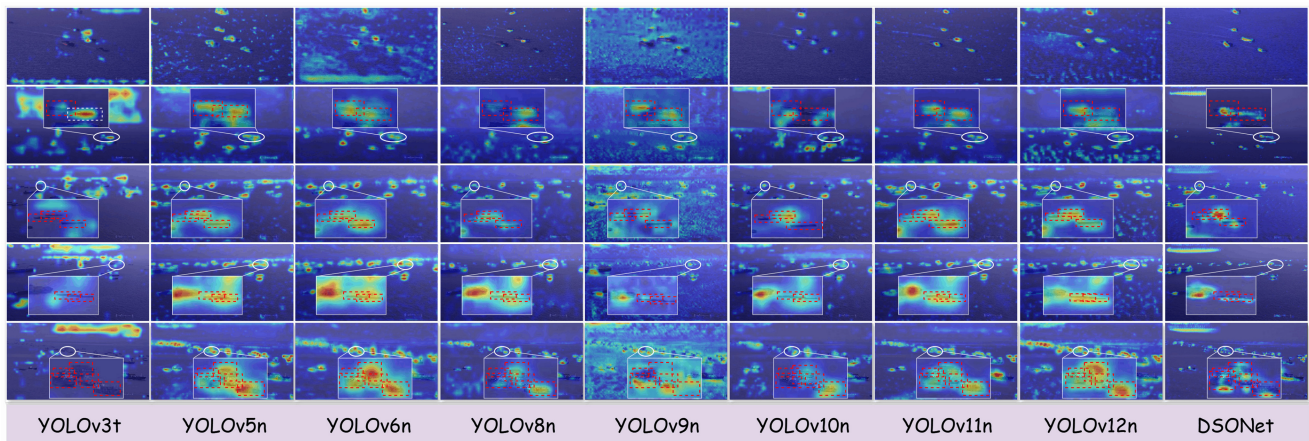


Fig. 11. Heatmaps for different degrees of occlusion. This shows detection results of other models and the proposed DSONet under conditions ranging from Unobstructed to Completely Obstructed. Input images with objects are overlaid with detection outputs. Ground truth is indicated by red dashed lines.

fusion and spatial recovery, especially for small and occluded ships, surpassing conventional upsampling techniques.

While DSONet reduces parameter complexity, we observe an increase in FLOPs due to its advanced feature processing design. The DFC module minimizes channel-wise computation but introduces a two-stage feature generation process that increases arithmetic operations. Similarly, AU performs dynamic spatial reconstruction using learned kernels, offering better accuracy at the cost of higher computation. The addition of the P2 layer for small-object detection and multiple concatenation operations in the feature pyramid also contribute to FLOPs growth. Overall, this tradeoff reflects a deliberate shift from redundancy to precision: higher computation is exchanged for improved detection performance, particularly in challenging maritime environments.

Despite its strengths, DSONet has limitations in extreme occlusion scenarios, especially when over 70% of a target is obscured. In such cases, feature degradation can lead to false positives or missed detections. Future work will explore incorporating attention mechanisms to enhance spatial focus and robustness under severe occlusion.

VI. CONCLUSION

In this article, we propose DSONet, a lightweight detection framework optimized for multiscale ship occlusion scenarios. By integrating the task-specific DF-Det architecture and the AU module, DSONet effectively improves feature extraction efficiency and spatial recovery for small and occluded targets while maintaining low computational cost. The experimental results validated its superiority over existing methods in both accuracy and robustness. In future work, we aim to extend DSONet to broader detection tasks and datasets within the maritime domain.

REFERENCES

- [1] L. P. Perera, P. Oliveira, and C. G. Soares, "Maritime traffic monitoring based on vessel detection, tracking, state estimation, and trajectory prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 13, no. 3, pp. 1188–1200, Sep. 2012.
- [2] A. Felski and K. Zwolak, "The ocean-going autonomous ship—Challenges and threats," *J. Mar. Sci. Eng.*, vol. 8, no. 1, p. 41, 2020.
- [3] V. F. Arguedas, G. Pallotta, and M. Vespe, "Maritime traffic networks: From historical positioning data to unsupervised maritime traffic monitoring," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 3, pp. 722–732, Mar. 2018.
- [4] W. Xiong, P. Van Gelder, and K. Yang, "A decision support method for design and operationalization of search and rescue in maritime emergency," *Ocean Eng.*, vol. 207, Jul. 2020, Art. no. 107399.
- [5] G. Xiao, Y. Wang, R. Wu, J. Li, and Z. Cai, "Sustainable maritime transport: A review of intelligent shipping technology and green port construction applications," *J. Mar. Sci. Eng.*, vol. 12, no. 10, p. 1728, 2024.
- [6] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multi-scale object detection in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 3–22, Nov. 2018.
- [7] N. Wang, Y. Wang, Y. Feng, and Y. Wei, "AodeMar: Attention-aware occlusion detection of vessels for maritime autonomous surface ships," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 13584–13597, Oct. 2024.
- [8] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.
- [9] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2016, pp. 779–788.
- [10] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 21–37.
- [11] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 1–21.
- [12] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," in *Proc. 38th Conf. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 107984–108011.
- [13] M. Tan, R. Pang, and Q. V. Le, "Efficientdet: Scalable and efficient object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 10781–10790.
- [14] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
- [15] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.
- [16] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
- [17] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 16965–16974.
- [18] T. Liang et al., "CBNet: A composite backbone network architecture for object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 6893–6906, 2022.
- [19] Y. Cao, K. Chen, C. C. Loy, and D. Lin, "Prime sample attention in object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 11583–11591.

- [20] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.
- [21] J. B. Campbell and R. H. Wynne, *Introduction to Remote Sensing*. New York, NY, USA: Guilford Press, 2011.
- [22] T. Li, H. Chang, M. Wang, B. Ni, R. Hong, and S. Yan, "Crowded scene analysis: A survey," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 3, pp. 367–386, Mar. 2015.
- [23] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, 2021.
- [24] Z. Zhang, L. Zhang, Y. Wang, P. Feng, and R. He, "ShipRSImageNet: A large-scale fine-grained dataset for ship detection in high-resolution optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8458–8472, 2021.
- [25] Z. Shao, W. Wu, Z. Wang, W. Du, and C. Li, "Seaships: A large-scale precisely annotated dataset for ship detection," *IEEE Trans. Multimedia*, vol. 20, no. 10, pp. 2593–2604, Oct. 2018.
- [26] Y. Chang, H. Chen, F. Wang, C. Chen, and W. Zeng, "MID: A comprehensive shore-based Dataset for multi-scale dense ship occlusion and interaction scenarios," 2024, *arXiv:2412.05871*.
- [27] C. Chen et al., "RSDS: A Specialized loss calculation method for dense small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Oct. 2024, Art. no. 4211817.
- [28] S. Liu et al., "Multi-scale ship detection algorithm based on a lightweight neural network for spaceborne SAR images," *Remote Sens.*, vol. 14, no. 5, p. 1149, 2022.
- [29] Y. Zhang, C. Chen, R. Hu, and Y. Yu, "ESarDet: An efficient SAR ship detection method based on context information and large effective receptive field," *Remote Sens.*, vol. 15, no. 12, p. 3018, 2023.
- [30] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
- [31] Z. Dai, H. Liu, Q. V. Le, and M. Tan, "CoatNet: Marrying convolution and attention for all data sizes," in *Proc. 35th Conf. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 3965–3977.
- [32] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 10012–10022.
- [33] L. Qi, L. Jiang, S. Liu, X. Shen, and J. Jia, "Amodal instance segmentation with kins dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2019, pp. 3014–3023.
- [34] S. Chen, P. Sun, Y. Song, and P. Luo, "DiffusionDet: Diffusion model for object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 19830–19843.
- [35] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
- [36] A. G. Howard et al., "Mobilenets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.
- [37] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "Ghostnet: More features from cheap operations," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 1580–1589.
- [38] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
- [39] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
- [40] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
- [41] C. Li, A. Zhou, and A. Yao, "Omni-dimensional dynamic convolution," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–20. [Online]. Available: <https://openreview.net/forum?id=DmpCfq6Mg39>
- [42] D. Qin et al., "MobileNetV4: Universal models for the mobile ecosystem," in *Proc. Eur. Conf. Comput. Vis.*, pp. 78–96, 2024.
- [43] X. Ma, X. Dai, Y. Bai, Y. Wang, and Y. Fu, "Rewrite the stars," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 5694–5703.
- [44] M. Lei et al., "YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception," 2025, *arXiv:2506.17733*.
- [45] L. Bai, C. Yao, Z. Ye, D. Xue, X. Lin, and M. Hui, "Feature enhancement pyramid and shallow feature reconstruction network for SAR ship detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1042–1056, 2023.
- [46] T. Zhang et al., "Balance learning for ship detection from synthetic aperture radar remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 190–207, Dec. 2021.
- [47] C. Chen, W. Zeng, X. Zhang, and Y. Zhou, "CS²Net: A remote sensing detection network breaking the second-order limitation of transformers with recursive convolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Sep. 2023, Art. no. 4207315.
- [48] H. Chen, C. Chen, F. Wang, Y. Shi, and W. Zeng, "RSNet: A light framework for the detection of multi-scale remote sensing targets," 2014, *arXiv:2410.23073v1*.

Chang Qu received the bachelor's degree in Internet of Things engineering from Dalian Minzu University, Dalian, China, in 2024. She is currently pursuing the master's degree in computer technology with Shanghai Maritime University, Shanghai, China.

Her research interests mainly include deep learning and object detection.

Yuhu Shi received the Ph.D. degree in information management and system from Shanghai Maritime University, Shanghai, China, in 2018.

He is currently an Associate Professor with the Information Engineering College, Shanghai Maritime University, Shanghai, China. His research interest is on artificial intelligence and machine learning, image processing and pattern recognition, as well as neural computing and biological information mining.

Hongyu Chen received the B.S. degree in computer science and technology from Nanjing Tech University, Nanjing, China, in 2023. He is currently pursuing the M.S. degree in computer science and technology with Shanghai Maritime University, Shanghai, China.

He has authored some academic articles which have been published in refereed international journals, such as *Neural Networks*, *IEEE TRANSACTIONS ON MEDICAL IMAGING*, and *IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT*. His research interests include machine learning, computer vision, high performance computing, and natural language processing.

Qianqian Ye received the bachelor's degree in software engineering from City College, Wuhan University of Science and Technology, Wuhan, China, in 2024. She is currently pursuing the master's degree with the School of Information Engineering, Shanghai Maritime University, Shanghai, China.

Her research interests mainly include low-light object detection and image segmentation.

Yang Cai received the bachelor's degree in software engineering from Nanjing Vocational University of Industry Technology, Nanjing, China, in 2024. He is currently pursuing the M.S. degree with the College of Information Engineering, Shanghai Maritime University, Shanghai, China.

His research interests mainly include object tracking and deep learning.

Yugang Chang received the bachelor's degree in naval architecture and marine engineering from Jimei University, Xiamen, Fujian, China, in 2016. He is currently pursuing the Ph.D. degree with the Merchant Marine College, Shanghai Maritime University, Shanghai, China.

His research interests mainly include object tracking, small object detection, and intelligent transportation systems.

Chengcheng Chen (Graduate Student Member, IEEE) received the bachelor's degree in computer science and technology from Jining University, Qufu, Shandong, China, in 2020. He is currently pursuing the Ph.D. degree with the Merchant Marine College, Shanghai Maritime University, Shanghai, China.

His research interests mainly include small object detection, semantic segmentation, and object tracking.

Fei Wang received the bachelor's degree in computer science and technology from Shanghai Polytechnic University, Shanghai, China, in 2023. He is currently pursuing the master's degree with the College of Information Engineering, Shanghai Maritime University, Shanghai.

His research interests mainly include deep learning and object detection.

Weiming Zeng (Senior Member, IEEE) received the Ph.D. degree in biomedical engineering from Southeast University, Nanjing, China, in 2005.

He is currently a Full Professor with the College of Information Engineering, Shanghai Maritime University, Shanghai, China. His research interests include image processing and analyzing, functional connectivity, and neuroscience computation.